

RED SEAL EXAM PREP

420B Ironworker (Generalist)

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings
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Mock Exam — answer all questions, then check the key

Q1. What is the purpose of "stirrups" or "ties" in a reinforced concrete beam or column?

- A) To position the longitudinal bars at the correct concrete cover depth
- B) To provide anchorage for the formwork
- C) To resist shear and confine the concrete core
- D) To resist longitudinal tension forces along the beam axis

Q2. An ironworker is welding a moment connection on a seismic force-resisting frame. What additional requirement applies compared to a regular gravity connection?

- A) All welds must be made using GTAW process
- B) The connection must be welded vertically-up only
- C) Higher amperage settings must be used to achieve deeper penetration
- D) Notch-tough weld metal with CVN requirements

Q3. On a structural drawing, what does the note "FSBW" refer to?

- A) Field Shop Bolt Work
- B) Full-Strength Butt Weld
- C) Fillet Strength Below Weld line
- D) Fabricated Steel Bearing Wall

Q4. An ironworker is about to work on fall arrest at the top of a column. What must the employer have in place before that system is used at all?

- A) Written rescue procedures for a worker whose fall is arrested
- B) A second worker at the base ready to lower the crane's hook
- C) A signed waiver acknowledging the risk of suspension trauma
- D) A spare harness and lanyard staged at grade for the rescuer

Q5. Synthetic web slings must be removed from service when:

- A) There are cuts, tears, or chemical burns visible
- B) The sling has been used for more than 500 lifts
- C) Only when the rated capacity tag is illegible
- D) Colour fading or minor surface abrasion is visible

Q6. Standard rebar ties are made using:

- A) Epoxy adhesive at each bar crossing
- B) Thin plastic zip ties rated for 50 kg
- C) 16-16.5 gauge soft annealed steel wire
- D) Spot welding at each intersection

Q7. A column is being lowered onto its anchor rods. What is the correct practice for the connectors receiving it?

- A) Steady the base plate by hand so the holes drop over the rods
- B) Guide it with taglines, hands and feet clear of the base plate
- C) Foot the base plate from the footing to stop it from swinging
- D) Slack the load line so the column settles onto the rods freely

Q8. What does a "Personal Fall Arrest System" (PFAS) consist of?

- A) Safety net and horizontal lifeline only
- B) Safety boots, hard hat, and safety glasses
- C) Body belt, rope grab, and vertical lifeline
- D) Full body harness, connecting lanyard, and anchor point

Q9. What does the abbreviation "TOS" mean on a structural steel drawing?

- A) Top Of Steel — the elevation of the top surface of the steel beam
- B) Total Overall Span — the beam span between columns
- C) Tension On Steel — a notation for tension member design
- D) Type Of Section — indicating the steel section type

Q10. A rigid steel weldment is picked on a four-leg bridle sling. Under British Columbia's OHS Regulation, the working load limit of the whole sling is limited to the working load limit of how many of its legs?

- A) Two of them, the pair on the same diagonal
- B) Whatever number a professional engineer sets
- C) All four, since the four legs share the load
- D) Three of them, whichever three are chosen

Q11. What does a "flag" on a welding symbol indicate?

- A) The weld is to be made in the field (field weld)
- B) All welds on the drawing must be made in this location
- C) The weld must be inspected by UT (ultrasonic testing)
- D) The weld is a complete joint penetration weld

Q12. Direct Tension Indicators (DTIs) are small washers with raised bumps. A connection is properly pretensioned when:

- A) All bumps on the DTI are still fully raised
- B) The bumps compress to a specified feeler-gauge gap
- C) The DTI is cracked and the bumps disappear
- D) The bolt head has been rotated exactly 1/2 turn from snug-tight

Q13. On a structural drawing, the connection dimensions for a diagonal brace are run from a mark labelled W.P., the working point. Unless the detail states otherwise, what does that mark locate on the joint?

- A) The corner of the gusset plate nearest the column face
- B) The point where the brace edge crosses the beam top flange
- C) Where the centrelines of the connected members intersect
- D) The centre of the bolt group joining the brace to the gusset

Q14. An anchor rod at a column base is 25 mm out of position. Which statement about correcting it in the field is correct?

- A) Reaming the base plate hole oversize keeps the rod's capacity
- B) Cutting it off and welding on an extension is a standard field fix
- C) Bending it cold with a rod bender restores its design strength
- D) Heating or flame-cutting the rod changes its strength properties

Q15. Tension control (TC) bolts are identified by a splined end. The installation is complete when:

- A) The splined end shears off during installation
- B) The direct tension indicator gap measures zero
- C) The bolt has been torqued to the specified value with a calibrated torque wrench
- D) The bolt head has been rotated 1/2 turn from snug-tight

Q16. Under AWS D1.1, when welding ASTM A572 Grade 50 steel with low-hydrogen SMAW electrodes, the minimum prequalified preheat and interpass temperature for material over 38 mm up to 65 mm thick is approximately:

- A) 107°C (225°F)
- B) No preheat required
- C) 66°C (150°F)
- D) 10°C (50°F)

Q17. What does "100% tie-off" mean for ironworkers during structural steel erection?

- A) The ironworker must be tied off for 100% of the workday
- B) 100% of the rigging must be secured before erection begins
- C) The ironworker must use a full 100-foot lifeline at all times
- D) At least one PFAS connection at all times, even when moving

Q18. What is a spreader beam used for in rigging?

- A) To measure the balance point of a long structural member
- B) To spread two crane boom sections for tandem lifts
- C) To counterbalance the weight of the crane boom
- D) To distribute the load over several pick points

Q19. What does the term "camber" mean when applied to a structural steel beam?

- A) A pre-set upward bow built into the beam to compensate for deflection under load
- B) The horizontal curvature of the beam due to lateral wind load
- C) The vertical angle of the beam connection at the column
- D) The distance between the top and bottom flanges of the beam

Q20. What is the minimum number of bolts that must be installed before erection bolts can be removed from a connection?

- A) At least one bolt installed in the connection
- B) At least two bolts or 25% of the bolt count, whichever is greater
- C) All bolts must be snug-tight before erection bolts can be removed
- D) At least 50% of the total bolt count in the connection

Q21. What does CSA W47.1 govern?

- A) The testing and inspection of completed structural welds
- B) The certification of fusion welding companies in Canada
- C) The classification of welding consumables
- D) The design of structural steel connections

Q22. A weld inspection reveals "undercut" along the toe of a fillet weld. What caused this?

- A) Excessive amperage or travel speed
- B) Insufficient preheat temperature
- C) Electrode too small for the fillet size
- D) Moisture in the electrode coating

Q23. Who is responsible for ensuring a lift plan exists for a critical lift?

- A) The rigging foreman and ironworker in charge
- B) The certified crane operator only
- C) The crane manufacturer's representative
- D) The project's structural engineer only

Q24. What is the purpose of a base plate in a steel column installation?

- A) To prevent corrosion of the column base
- B) To hold the anchor bolts
- C) To allow lateral adjustment after erection
- D) To spread the column load over the foundation

Q25. In structural steel work, what does CSA W59 govern?

- A) The certification of individual welders for structural work
- B) The testing and classification of welding consumables
- C) The safety requirements for welding fumes and gases
- D) The design and fabrication of welded steel structures

Q26. What is the effective throat of a 12 mm fillet weld?

- A) 6 mm
- B) 8.5 mm
- C) 10 mm
- D) 12 mm

Q27. When installing high-strength bolts using the turn-of-nut method, what is the first step?

- A) Install all bolts finger-tight only
- B) Apply lubricant to all bolt threads
- C) Bring all bolts to snug-tight first
- D) Apply full final torque immediately

Q28. Which crane hand signal means "STOP"?

- A) Arm extended, palm facing down, moving arm side to side
- B) Both arms raised above head
- C) One arm raised, rotating at the elbow
- D) Arm extended, palm facing the crane operator, fist closed

Q29. Rebar chairs (bar supports) are used to:

- A) Hold rebar at the correct height for cover
- B) Hold vertical rebar in position while tying
- C) Mark the location of rebar on the concrete surface after pouring
- D) Splice rebar together at joints

Q30. Rebar arrives on site carrying tight mill scale and light surface rust. What should the ironworker do before it is placed?

- A) Apply a rust inhibitor coating before placing in forms
- B) Place the bars as is — light rust is acceptable
- C) Replace all lightly rusted bars with new material
- D) The bars must be sandblasted clean before placement

Q31. An ironworker inspects a 6×19 IWRC wire rope sling before a critical lift and finds 11 randomly distributed broken wires in one rope lay. What action should be taken?

- A) Remove the sling from service immediately
- B) The sling may be used if the broken wires are not load-bearing
- C) Tag the sling for monitoring and continued use
- D) The sling may be used — the limit is 20 broken wires per rope lay

Q32. On an Ontario steel erection project, a safety net used as a fall protection system must be designed, tested, and installed in accordance with:

- A) CSA Z259.10, the CSA standard that covers full body harnesses
- B) OSHA 1926.502, which Canada adopts by reference for steel erection
- C) The net manufacturer's instructions alone, with no referenced standard
- D) ANSI/ASSE A10.11, adopted by Ontario's construction regulation

Q33. What is a "beam clamp" used for in ironworking?

- A) To permanently connect two beams at a splice
- B) To attach rigging to a beam flange without welding
- C) To clamp the welding electrode holder to the beam during welding
- D) To hold a beam in plumb during erection until it is bolted

Q34. Under the Alberta OHS Code, what is the maximum free-fall distance permitted for a personal fall arrest system that has no shock absorber?

- A) 0.6 m (2 ft)
- B) 3 m (10 ft)
- C) 1.2 m (4 ft)
- D) 1.8 m (6 ft)

Q35. During a crane lift, what is the purpose of a tagline?

- A) To control the rotation and swing of the load
- B) To measure the length of the lift path
- C) To attach the load to the crane hook
- D) To provide an extra rigging connection in case the main sling fails

Q36. A wire rope sling is rated for a working load limit (WLL) of 10,000 kg in a straight-line (vertical) hitch, and the sling tag shows no separate choker rating. What is the effective WLL in a choker hitch at an angle of choke of 120° or greater?

- A) 5,000 kg — choker hitch reduces capacity by 50%
- B) 10,000 kg — choker hitch does not reduce capacity
- C) 8,000 kg — choker hitch reduces capacity by 20%
- D) 7,500 kg — choker hitch reduces capacity by 25%

Q37. What do the deformations (ribs) on deformed rebar do?

- A) Reduce the bar cross-section to save material at low-stress locations
- B) Mechanically interlock with concrete to improve bond and force transfer
- C) Increase the bar weight to improve resistance to impact loads
- D) Provide identification markings for the steel mill

Q38. On a structural drawing, what does the symbol "EL. 12500" typically mean?

- A) The point's elevation above the project datum
- B) Element number 12,500 on the fabrication list
- C) The element has a design load of 12,500 kg
- D) The element is 12,500 mm in overall length

Q39. A beam has been landed in its connection and the connectors are ready to cut it loose from the crane. What has to be in that connection at that moment?

- A) Two bolts run on by hand so the beam can still be shifted
- B) Enough bolts to stop rotation, drawn up wrench-tight
- C) One bolt wrench-tight and a drift pin in a second hole
- D) Every bolt run in and torqued to the design drawings

Q40. Before using any rigging hardware (shackles, hooks, chains) on a job site, the ironworker should:

- A) The equipment only requires inspection during the annual company audit
- B) Visually inspect for damage and verify adequate WLL
- C) Test each piece by applying twice the WLL to check for deformation
- D) Apply a fresh coat of oil to all moving parts

Q41. No manufacturer's specification can be found for a winding drum that a wire rope will spool onto. What minimum drum diameter should be used?

- A) 20 times the rope diameter
- B) 6 times the rope diameter
- C) 30 times the rope diameter
- D) 12 times the rope diameter

Q42. What is a "lap splice" in reinforcing steel?

- A) A welded joint fusing two rebar pieces at their ends
- B) A mechanical coupler that joins two rebar end-to-end
- C) An overlap where two parallel bars transfer stress
- D) A bar bent into a U-shape hooked around another bar

Q43. A structural drawing shows "4-M20 A325 bolts" at a beam-to-column shear tab connection. What does this mean?

- A) A bolt group with a shear capacity of 4,000 N per M20 bolt
- B) Metric bolts, Group 4 size class, rated 20 kN shear capacity each
- C) 4 machine bolts of 20 mm diameter, Grade A325 galvanized finish
- D) 4 bolts, metric M20 (20 mm diameter), ASTM A325 high-strength

Q44. A complete joint penetration (CJP) groove weld requires:

- A) Weld metal to fill the groove at least halfway
- B) Post-weld heat treatment regardless of steel grade
- C) Weld metal penetrating the joint's full thickness
- D) At least three weld passes in the groove

Q45. An opening is left in a steel deck for a duct and, instead of a guardrail, the crew covers it. Under Ontario's Construction Projects Regulation, which of these must that covering satisfy?

- A) It is inspected and certified by an engineer before use
- B) It is used only where a guardrail cannot be installed
- C) It is left liftable so materials can be passed through
- D) It is fastened so it cannot be moved off the opening

Q46. A hydraulic bolt tensioner is used instead of a torque wrench for high-strength bolt installation. What is the key advantage?

- A) It works on all bolt grades including stainless steel and aluminum
- B) It applies direct tension, eliminating torque variability
- C) It is lighter and cheaper than a torque wrench
- D) It can install bolts in confined spaces where a torque wrench cannot fit

Q47. A beam detailed with 25 mm of camber arrives on site with its camber paint mark worn off. How does the crew establish which way up it goes before it is flown?

- A) Sight along the beam on the ground to find its natural bow
- B) Measure the depth at both ends and put the deeper end up
- C) Fly it either way, since 25 mm is inside erection tolerance
- D) Assume the mill mark stamped on the web is meant to face up

Q48. Steel floor decking should be laid in what sequence?

- A) Outward from the crane or access side
- B) Starting at the column and working toward the beam midspan
- C) In whatever order suits the crew that shift
- D) Starting from the centre of the bay and working to the edges

Q49. A crane is lifting a 20,000 kg load at a 15m radius. The load chart shows maximum lift capacity at 15m is 18,000 kg (with all outriggers fully extended on firm ground). What should the ironworker signalperson communicate to the operator?

- A) Proceed if the crane manufacturer approves in writing
- B) Proceed if the lift will be completed in under 2 minutes
- C) Proceed — crane charts include a 10% safety margin for occasional overloads
- D) Stop the lift — it exceeds the chart capacity

Q50. An optical level or builder's level is used during structural steel erection to:

- A) Establish level reference elevations across the structure
- B) Read the crane load chart to determine crane capacity
- C) Measure the plumb of columns using the level bubble
- D) Check the alignment of bolt holes in connections

Answer Key & Explanations

Q1. Answer: C

Stirrups (beams) and ties (columns) resist shear and torsion forces, confine the concrete core to increase ductility, and brace the longitudinal bars against buckling under compression load.

Q2. Answer: D

Seismic connections take large cyclic deformations and have to absorb that energy by yielding rather than by fracturing. The weld metal must therefore be notch tough, which is proven by Charpy V-notch (CVN) impact testing at the low temperature and minimum absorbed energy called up by the project specification and the accepted welding procedure. For the same reason the procedure normally sets a maximum heat input and a maximum interpass temperature, because too much heat coarsens the grain and costs the weld its toughness. In Canada the design of seismic force-resisting systems is governed by CSA S16, referenced through the National Building Code; the American equivalents, AISC 341 and AISC 358, apply only on a project whose specification calls them up.

Q3. Answer: B

FSBW (Full Strength Butt Weld) or CJP (Complete Joint Penetration) groove weld indicates that the weld must develop the full tensile strength of the connected base metal — the weld is not the weak link. This requires 100% penetration through the joint thickness, usually verified by visual inspection plus UT (ultrasonic testing) or radiography. It is different from: a partial joint penetration (PJP) weld, which does not penetrate the full thickness and has reduced capacity; a fillet weld, which joins members at a surface (no groove); and a plug or slot weld. FSBW/CJP welds are required at moment connections (beam-column moment joints) and splice plates carrying tension — they cost more (more prep, more passes, more inspection) than fillet welds.

Q4. Answer: A

Ontario Regulation 213/91 s. 26.1 (4): "Before any use of a fall arrest system or a safety net by a worker at a project, the worker's employer shall develop written procedures for rescuing the worker after his or her fall has been arrested." The requirement is a precondition, not an afterthought - it bites before any use of the system, so a crew that ties off without a rescue plan is offside from the first connection. The reason is that arresting the fall is only half the event. A worker hanging in a harness is held by leg straps that restrict blood return, and suspension becomes dangerous in a time frame measured in minutes, so a rescue that has to be invented on the spot is a rescue that arrives late. Staging spare equipment, or posting someone at the base, may well form part of a procedure, but neither is the requirement and neither is any use without a plan that says who does what, with what, and how fast. A waiver is not a control and does not discharge a duty under the regulation.

Q5. Answer: A

A synthetic web sling comes out of service for damage that has cut, burned, melted or abraded the fibres or the stitching. Alberta's OHS Code s.305(2) requires permanent removal if part of the sling is melted, charred or damaged by chemicals, if stitches in load-bearing splices are broken or worn, or if end fittings are excessively pitted or corroded, cracked, distorted or broken. Section 305(1) adds the measured limits: an edge cut longer than the web thickness, or abrasion deeper than 15 percent of the webbing thickness taken across all plies. Colour fading and light surface scuffing are not by themselves removal criteria. No regulation retires a sling on a lift count. An illegible or missing capacity tag does require removal, but it is one criterion among many rather than the only one.

Q6. Answer: C

Rebar ties are made from 16 to 16.5 gauge (approximately 1.6 mm diameter) smooth, soft annealed steel wire, twisted with a tier tool. Multiple tie patterns exist for different conditions (snap tie, saddle tie, figure-8 tie).

Q7. Answer: B

IHSA's Construction Health and Safety Manual chapter on structural steel erection (Ironwork, M036, 2023 edition) gives two controls for column standing: keep the load line vertical and taut while the column is being stood, and keep hands and feet clear of the base plate while the column is being lowered onto the anchor rods. The base plate is the pinch point. A column landing on four rods can shift sideways in an instant, and a hand steadying the plate or a boot bracing it against the footing is between a plate of several hundred kilograms and hard concrete. Control the piece from a distance with taglines instead. Slackening the load line before the column is stable is the other error in this list: the same chapter directs that the column be stable before the rigging is disconnected, and a slack line lets the piece take charge of itself while the crew is still reaching into the connection.

Q8. Answer: D

A PFAS has three essential components: (1) full-body harness, (2) connecting means (lanyard, self-retracting lifeline, rope grab + lifeline), and (3) an anchor point with sufficient strength.

Q9. Answer: A

TOS (Top of Steel) is an elevation reference on structural drawings indicating the height of the top flange of a beam, measured from the project datum. It is used to coordinate with the concrete deck, mechanical, and architectural drawings. Common related notations: BOS (Bottom of Steel — soffit elevation, important for clearance), TOS is NOT the same as finished floor elevation (which is TOS + deck slab thickness + topping if any). Wrong answers often confuse TOS with the beam depth or with the centreline elevation. For steel erection: when setting beam elevation, you level to the TOS — not to the bottom flange or web centreline.

Q10. Answer: D

WorkSafeBC's OHS Regulation s. 15.33 (2) is one sentence: the working load limit of a sling with more than three legs is limited to the working load limit of any three legs of the sling. A fourth leg is allowed on the hook; it simply cannot be counted. The mechanics behind that are worth carrying to any jurisdiction. Four legs holding one rigid piece are statically indeterminate - there are more supports than the statics of the lift can define - so which legs are actually tight is decided by millimetres of leg length and by exactly where the pick points sit, not by symmetry. Nothing about the arrangement guarantees that all four come up tight together, so adding four leg capacities describes a lift that may not happen. The section does not stop at two, though: three is the cap it states. An engineer's stamp is the answer people reach for when a rule looks severe, but this subsection sets the cap itself and does not hand the number to anyone to reset. Two further limits ride along in the same section: the working load limit of no individual component of the assembly may be exceeded (15.33 (1)), and the load carried by any single leg of a bridle must not exceed the working load limit of that leg (15.33 (3)).

Q11. Answer: A

A flag on the weld symbol reference line indicates a field weld — one that is to be made during erection at the job site, as opposed to a shop weld made during fabrication.

Q12. Answer: B

DTIs work by compressing their raised protrusions as bolt pretension increases. When sufficient tension is reached, the gaps between the protrusions and the contact surface fall below the specified maximum refusal value, verified with a feeler gauge.

Q13. Answer: C

The working point is where the centrelines - the gravity axes - of the members meeting at that joint intersect.

The detailer sets the joint out from that point so that the lines of action of the column, the beam and the diagonal brace all meet in one place. When they do, the joint delivers axial force and no member picks up a bending moment from an offset. Shift the brace's line of action away from the working point and an eccentricity is introduced, and the moment that eccentricity produces has to be carried by the connection and by the members framing into it. The choice of working point belongs to the engineer, so a detail that deliberately works to a point away from the member centrelines says so on the drawing and the connection is designed for that eccentricity - which is why the stem says "unless the detail states otherwise".

For the crew the working point is a layout reference, not a piece of hardware. On a brace-to-beam-to-column joint it commonly falls where the column and beam centrelines cross, inside the members themselves, where there is nothing to fasten to; the gusset plate exists precisely because the members cannot all physically meet at that point. That is what separates it from the three wrong answers, each of which is a real feature you can put a tape on. The centre of the bolt group is where the brace is fastened to the gusset, and its position is set out from the working point rather than the other way round. The corner of the gusset and the place where the edge of the brace crosses the beam flange are edges of fabricated plate and shape; they move if the gusset is resized or the brace section is changed, while the working point stays where the member centrelines cross.

Q14. Answer: D

Heat is the thing to be careful of. An anchor rod's strength comes from its steel chemistry and, in the higher grades, from heat treatment; taking a torch to it drives the material through temperatures that alter those properties, and flame-cutting leaves a heat-affected zone and a reduced section in a member designed to carry tension. Neither effect is visible afterward, which is what makes it dangerous - the rod looks fine and is no longer the rod the engineer designed. That is why heating and cutting are not the erector's call, and why any remedy for a misplaced rod comes back with engineering direction. The other three are the field fixes people reach for and none of them is sound. Cold bending does not restore design strength; it works the steel, leaves residual stress at the bend and puts a permanent kink where the load path wanted a straight bar. Welding an extension onto a cut rod assumes the rod steel is weldable and that a field splice can carry the design tension, and neither can be assumed. Reaming the plate hole oversize changes how load transfers between rod and plate and can leave the rod bearing on very little, which is why oversized holes at base plates carry their own plate washer requirements.

Q15. Answer: A

TC bolts have a breakneck groove at the splined end. When minimum pretension is achieved, the torque reaction causes the spline to shear off cleanly, providing visual verification that the required minimum pretension has been reached.

Q16. Answer: C

AWS D1.1 places ASTM A572 Grade 50 in Category B, and Category B welded with low-hydrogen SMAW electrodes calls for 150°F (66°C) on material over 38 mm through 65 mm thick. The full Category B row of the prequalified preheat table (Table 5.8 in the 2020 and later editions; Table 3.2 in earlier ones) reads: 3 to 20 mm = 32°F (0°C); over 20 through 38 mm = 50°F (10°C); over 38 through 65 mm = 150°F (66°C — the metric column of D1.1M rounds this to 65°C); over 65 mm = 225°F (107°C). So 107°C is correct only for plate thicker than 65 mm, and that is the trap in this question. Watch the band edges too: a section exactly 38 mm thick still falls in the 10°C band, because the row reads 'over 38'. On Canadian structural work AWS D1.1 is not the governing document. Building Code Article 4.3.4.1 requires buildings and structural members made of structural steel to conform to CAN/CSA S16, and S16 in turn requires a CSA W47.1 certified fabricator with joint design and quality to CSA W59 — so preheat comes off the CWB-accepted WPS/WPDS, never off a US table from memory. The Canadian counterpart grade is CSA G40.21 350W (350 MPa minimum yield, against 345 MPa for A572 Grade 50), which is what a Canadian mill certificate will normally show.

Q17. Answer: D

100% tie-off means continuous connection — the ironworker must have at least one PFAS connection at all times, including while moving or transitioning their tie-off point. A double-lanyard or SRL with a second connection allows transitioning without a gap.

Q18. Answer: D

A spreader beam is a below-the-hook lifting device that widens the pick points on a load. The beam carries the spread as compression along its own length, so the sling legs hang nearer to vertical and no horizontal squeeze is passed into the load itself. It does not counterbalance anything on the crane, it does not find a balance point, and it has nothing to do with boom sections. Do not confuse it with an equalizer beam, which is a different device: an equalizer beam pivots at a centre point to share load equally between two hoist lines or two cranes. Alberta's OHS Code s.297(2) puts below-the-hook lifting devices other than slings under ASME B30.20, and s.297(3) and (4) allow a spreader bar to be labelled with its rated capacity by a uniquely numbered capacity data sheet. BC's OHS Regulation requires the working load limit of a spreader bar to be certified by a professional engineer or established by the manufacturer (s.15.58), requires a nameplate (s.15.59), and counts the device as part of the lifted load (s.15.60).

Q19. Answer: A

Camber is a deliberate upward curve fabricated into a beam so that under design load, the beam deflects back to a level position. Common in long-span or heavily loaded beams.

Q20. Answer: B

Per AISC and CISC requirements, a minimum of two bolts or 25% of the bolt group (whichever is greater) must be installed and snug-tight before erection bolts are removed.

Q21. Answer: B

CSA W47.1 (Certification of Companies for Fusion Welding of Steel) governs the certification of welding companies — it certifies the company, not individual welders. Certified companies must have qualified welding supervisors, approved welding procedures (WPSs), and a documented quality control system. This is different from: CSA W59 (which governs the design and fabrication requirements for welded steel structures), CSA W178.2 (which certifies individual welding inspectors), and AWS D1.1 (the American Structural Welding Code used as a reference). On a Canadian structural steel project, the fabricator or erector must hold a current CSA W47.1 certification — this is a mandatory contract requirement.

Q22. Answer: A

Undercut is a groove melted into the base metal at the weld toe that is not filled by weld metal. It comes from too much heat at the toe — excessive amperage, too fast a travel speed, too long an arc, or an electrode angle that throws the arc onto the vertical member instead of into the corner. It matters because it removes section from the base metal and leaves a sharp notch at the toe, which is where fatigue cracks start. The other faults listed cause different problems: moisture in the coating causes porosity, too little preheat on heavy or restrained sections invites cracking, and too small an electrode gives an undersized weld rather than a gouged toe.

Q23. Answer: A

The rigging crew (foreman/competent rigger) and the ironworker in charge of the lift, in coordination with the site supervisor, are responsible for preparing or implementing the lift plan for critical lifts. The engineer may provide structural input.

Q24. Answer: D

A base plate distributes the concentrated column load over a larger bearing area on the concrete foundation, reducing bearing stress to within allowable limits.

Q25. Answer: D

CSA W59 (Welded Steel Construction — Metal Arc Welding) specifies design, fabrication and erection requirements for welded steel structures in Canada, including joint design, acceptable workmanship and inspection criteria. It is often confused with CSA W47.1. W59 is the construction standard — what must be built and to what quality — while W47.1 certifies a fabricator or erector as a company qualified to do the welding, and CSA W178.2 certifies welding inspectors. Welding consumables belong to the CSA W48 series and welding health and safety to CSA W117.2, so neither of those sits in W59 either. AWS D1.1 is the equivalent American standard; Canadian work is built to W59, and D1.1 governs only where a project specification calls it up. For the Red Seal ironworker exam: W59 is the welded construction standard, W47.1 is company certification, W178.2 is inspector certification.

Q26. Answer: B

For a standard fillet weld with equal legs, the theoretical throat = $\text{leg} \times \sin(45^\circ) = 12 \times 0.707 \approx 8.5 \text{ mm}$. The effective throat is used to calculate weld strength.

Q27. Answer: C

Turn-of-nut requires all bolts in the connection to be brought to the snug-tight condition first (full effort of an ironworker with a spud wrench), then a specified rotation is applied to achieve the minimum pretension.

Q28. Answer: A

The universal crane signal for STOP is arm extended horizontally with palm facing down, moved side to side. EMERGENCY STOP uses both arms in this position.

Q29. Answer: A

Bar chairs and supports are plastic or metal supports placed under rebar to maintain the required concrete cover by holding the bars at the correct height above the form (distance from bottom of rebar to bottom of slab).

Q30. Answer: B

Tight mill scale and light surface rust are acceptable and need not be removed; light rust may slightly improve bond by roughening the surface. Only loose scale and loose, flaking rust have to come off, and the bar must still meet its specified dimensions, mass and deformation height after hand wire brushing. Mill scale and rust are not the same thing: mill scale is the tight blue-black oxide left by hot rolling, while rust is the orange-brown corrosion product that forms afterwards. Canadian project specifications written to CSA A23.1 call for removal of loose scale and loose rust only.

Q31. Answer: A

ASME B30.9, para. 9-2.9.5(b)(1): a strand-laid or single-part wire rope sling must be removed from service at 10 randomly distributed broken wires in one rope lay, or 5 broken wires in one strand in one rope lay. Eleven randomly distributed breaks exceeds that limit, so the sling comes out of service immediately — tag it, segregate it so it cannot be picked up again, and report to the rigger-in-charge. Note what the standard does not say: B30.9 sets no broken-wire limit keyed to rope construction, so there is no separate 6×19 or 6×36 figure. Classification governs the minimum clear length of rope between splices (para. 9-2.3.2), not the wire count. The one construction that changes the count is the cable-laid sling, at 20 broken wires per lay. B30.9 is not merely an American reference here — it is adopted by regulation: Alberta OHS Code Part 21, s.297(1) requires wire rope slings manufactured on or after July 1, 2009 to meet ASME B30.9, and s.306 adds further removal triggers, including wear or corrosion affecting individual wires over more than one-third of the original rope diameter, bird-caging, kinking, and heat or arc damage. Beware of carrying figures across: provincial limits for running rope are different and tighter (Alberta s.306(2): 6 randomly distributed broken wires in one rope lay, or 3 in one strand), but a sling is not a running rope. Check the rigging part of your own province's regulation.

Q32. Answer: D

Ontario Regulation 213/91 (Construction Projects) s. 26.8 (1) requires a safety net to be designed, tested and installed in accordance with ANSI/ASSE Standard A10.11-2010, Safety Requirements for Personnel and Debris Nets. The rest of the section is worth knowing subsection by subsection, because each one lands on a different person. Under s. 26.8 (2) the net is installed by a competent worker. Under s. 26.8 (3) an engineer, or a competent person under the engineer's supervision, inspects and tests the installation before it is put in service. Under s. 26.8 (4) the engineer documents that inspection and testing. Under s. 26.8 (5) a copy of the document is kept at the project while the net is in service. OSHA 1926.502 is a United States regulation and has no legal force in Canada. The CSA Z259 standards listed in s. 26.1 (3) cover the parts of a personal system - Z259.10 is the full body harness standard - and none of them covers nets. Note also that it is the provincial regulation that decides which edition of the net standard applies and what else the net must do: Alberta's OHS Code s. 320 (1) (a) adopts the older ANSI A10.11-1989 (R1998) and then adds its own dimensions, requiring the net to extend not less than 2.4 metres beyond the work area (s. 320 (1) (d)) and not more than 6 metres below it (s. 320 (1) (e)). Ontario's s. 26.8, read through all five subsections, prints no such distance and leaves it to the referenced standard and to the engineer's inspection.

Q33. Answer: B

Beam clamps attach to a beam flange by clamping action, providing a temporary rigging point for hooks, slings, and blocks — for hoisting, pulling, or hanging equipment without welding, drilling, or otherwise permanently modifying the beam.

Q34. Answer: C

The Alberta OHS Code requires a personal fall arrest system without a shock absorber to limit a worker's free-fall distance to 1.2 m. A shock absorber is normally required in the first place; a system without one is allowed only where it meets that free-fall limit and the rest of the same section. Where a shock absorber is used, the permitted free fall is the maximum free-fall distance the absorber is labelled and rated for, and the system must still be arranged so the worker cannot strike the ground or a level below, and must limit the maximum arresting force on the worker to 6 kN, or 8 kN with an E6 shock absorber used to the manufacturer's specifications. The practical control on free fall is anchor height: the Code requires the lanyard to be secured to an anchor no lower than the worker's shoulder height, or as high as reasonably practicable.

Q35. Answer: A

Taglines allow workers to guide and control a suspended load from a safe distance, preventing rotation and swing without placing workers directly under or near the load.

Q36. Answer: D

ASME B30.9-2021, para. 9-2.10.1(d): when the choker hitch rating is not identified on the sling, the choker hitch rating for single-leg and bridle wire rope slings is 75% of the sling's straight-line hitch rating (70% for cable-laid slings). So $10,000 \text{ kg} \times 0.75 = 7,500 \text{ kg}$ — a 25% reduction, caused by the bending and pinching of the rope at the choke point. That 75% figure holds only at an angle of choke of 120° or greater: para. 9-2.10.1(e) sends you to Table 9-2.10.1-1 for smaller angles, which gives 87% of the choker rating at 90° – 120° , 74% at 60° – 89° , 62% at 30° – 59° , and 49% at 0° – 29° . Although B30.9 is an American standard, it governs Canadian work because occupational health and safety regulation adopts it by reference — Alberta OHS Code Part 21, s.297(1) requires wire rope, alloy steel chain, synthetic fibre rope, metal mesh and synthetic fibre slings manufactured on or after July 1, 2009 to meet ASME B30.9. If the sling tag carries its own choker rating, use the tag.

Q37. Answer: B

The transverse and longitudinal ribs on deformed bar create mechanical interlock with the surrounding concrete, dramatically improving bond strength compared to smooth bars.

Q38. Answer: A

EL. (Elevation) followed by a number — for example EL. 12500 — gives the finished height of that point in millimetres above the project datum, typically the finished first floor established at EL. 0. So EL. 12500 is 12,500 mm, or 12.5 m, above datum. Elevations above the datum are positive; elevations below it are negative. This is different from a grid coordinate (which locates a point in plan, not in height), a dimension (which measures the distance between two specific points), or a TOS/BOS notation (which gives the elevation of the top or bottom of a specific steel member). When setting anchor bolts or erecting columns, always verify the elevation against the structural drawings — not the architectural drawings, which may use a different datum.

Q39. Answer: B

IHSA's ironworker chapter puts it in one line: before being cut loose, the beam must be bolted so that it will not rotate, at least two or more bolts should be put in position depending on the engineer's requirements, and a drift pin or a wrench placed in a hole is not to be relied on. Two things therefore have to be true at that moment, and the count is only one of them. Enough fasteners have to be in to take rotation out of the connection, and IHSA's structural steel erection guide supplies the condition that goes with them: bolted and temporary connections are to be wrench-tight, and are torqued later, to the design specification, during bolt-up and detail work. Bolts run on by hand leave the beam free to move in the connection, which is the state the rule exists to prevent. A drift pin is a tool for drawing holes into line; it carries no load and can walk out, which is why the guidance names it as the thing not to trust. Waiting for the whole connection to be torqued keeps the crane tied to one beam - the bolting-up crew follows the erection crew and installs the remaining bolts afterward. How many bolts is not the connector's own call: the number comes from the engineer's requirements and the erection plan, and the same guide sets one bolt per connection for diagonal bracing unless the structural design engineer specifies otherwise, which is a different member and a different number.

Q40. Answer: B

Pre-use inspection of rigging hardware is mandatory. Visually inspect for cracks, deformation (hooks opening, shackle bowing), corrosion reducing cross-section, missing or illegible load ratings, and pin security — and verify the WLL is adequate for the lift.

Q41. Answer: A

The binding rule is the manufacturer's figure: Alberta's OHS Code s.302(1) requires that the wire ropes, sheaves, spools and drums used in rigging have a diameter of not less than the diameter specified by the manufacturer for that use. Where no specification can be found, Alberta's published explanation guide to Part 21 gives the working minimum — unless otherwise specified by the manufacturer, the diameter of winding drums should not be less than 20 times the diameter of the rope, which reduces the likelihood of weakened rope caused by excessive bending stresses. Six or twelve times the rope diameter bends the rope far harder than it is built to take; thirty times is generous but is not the published minimum. Bending fatigue, not overload, is what usually kills a rope. Note that there is no single number for a rope running over a sheave: the minimum varies with rope construction, so read the rope manufacturer's table rather than carrying one figure in your head.

Q42. Answer: C

A lap splice is an overlapping zone where two bars run parallel: the bar ends overlap by a specified length (development length $\times$ splice factor) so that bar forces transfer through bond with the surrounding concrete.

Q43. Answer: D

4-M20 A325 means 4 bolts of M20 size (20 mm nominal diameter) meeting high-strength structural bolt requirements. Since 2016 the governing product specification is ASTM F3125 — ASTM A325 and A325M were withdrawn and consolidated into F3125 as grades, so the metric bolt shown here is properly ASTM F3125 Grade A325M, the metric grade covering sizes M12 to M36. Head markings and drawing callouts were deliberately left unchanged, which is why the drawing and the bolt head still read A325 — an apprentice who goes looking for the standalone ASTM A325 standard will find a withdrawn document. On Canadian work the connection itself is governed by CSA S16, Design and construction of steel structures, which sets the resistance of the bolts and the connection and also the rules for installing and inspecting high-strength bolted joints, including pretensioned assemblies. Work to the clause numbering in the edition of that standard referenced by your provincial building code.

Q44. Answer: C

A CJP weld requires weld metal to completely penetrate the full thickness of the joint, filling the entire cross-section and achieving full fusion. It is required in joints subject to tension, cyclic, or seismic loading.

Q45. Answer: D

O. Reg. 213/91 s. 26.3 (2) gives two ways to stop a worker falling through an opening on a work surface - a guardrail system, or a protective covering - and it sets out what the covering has to be: it completely covers the opening, is securely fastened, is adequately identified as covering an opening, is made from material adequate to support all loads to which it may be subjected, and is capable of supporting a live load of at least 2.4 kilonewtons per square metre. Fastening is the one that site practice most often skips. A sheet lying loose over a hole is moved by the very traffic it is there to protect: kicked aside, slid off as material is dragged over it, or lifted by wind. The worker who then steps on the edge of a shifted cover goes through it.

That is also why a cover left deliberately liftable for access is not a compliant cover. Where work has to be done in or around the opening, s. 26.3 (3) allows the covering or the guardrail to be removed temporarily, but only while the worker is adequately protected and danger signs are posted in accordance with s. 44 (1) and (2) - a planned removal under protection, not a cover that anybody can pick up.

The other two answers borrow rules that belong elsewhere. Inspection and testing by an engineer, or by a competent person under the engineer's supervision, before the system goes into service is the safety net requirement in s. 26.8 (3); nothing in s. 26.3 puts an engineer on a deck cover. And the covering is not a fallback that first needs a guardrail to be impracticable: s. 26.3 (2) sets the two out as alternative precautions for an opening, so a covering that meets all five requirements is protection in its own right. Note also that this duty does not wait on a height - section 26 lists falling through an opening on a work surface as a hazard in itself, alongside falling more than 3 metres.

Q46. Answer: B

Hydraulic tensioners apply direct tensile load to the bolt by pulling the threaded end rather than applying torque, eliminating the torque-to-tension variability that comes from lubrication, thread condition, and technique. This achieves more consistent and accurate pretension.

Q47. Answer: A

Camber is a bow along the length of the beam, so it can be read off the piece itself: lay it on level blocking, sight down the top flange from one end, and the curve shows which way the piece rises at midspan. Twenty-five millimetres over a long span is easy to see once you sight it and nearly invisible if you only look at the beam side on. Check the reading against the shop detail for that piece mark before it leaves the ground, then re-mark it. The wrong answers each fail for a different reason. Depth at the ends says nothing about camber, because camber is put into a beam of constant depth by cold cambering or by heat, so both ends measure the same. The mill mark records the producer, the heat number and the grade; it carries no orientation meaning. And 25 mm is not a tolerance to be shrugged off: it is a deliberate quantity the designer put in to be cancelled by deflection under load, so a beam flown upside down starts 25 mm low and finishes about 50 mm out at midspan, which shows up in the floor and in the deck bearing.

Q48. Answer: A

Deck is laid working away from the crane or access side, so the crew always advances over deck that is already landed, supported and fastened instead of stepping onto loose sheets or out along bare beams. Beginning in the middle of a bay fails for the same reason - somebody has to cross open steel to get there - and letting the order suit the crew leaves loose sheets lying where people walk. Ontario's O. Reg. 213/91, s. 32 (1) sets the principle the sequence serves: during the construction of a building, temporary or permanent flooring shall be installed progressively as the building is erected. The fastening and loading requirements in s. 32 (2) are written for temporary flooring and are quoted only as such. The IHSA ironworkers' chapter adds the handling that goes with the sequence: bundles are landed on framing members so they are supported well enough to be un-banded without being dislodged, decking is secured so that wind cannot blow it off, and access to the floor where deck is being placed and to the area below it is limited to the workers doing that job.

Q49. Answer: D

Load chart values are maximums — they MUST NOT be exceeded under any circumstances. Crane load charts already include safety factors per CSA Z150. Exceeding the chart rating voids the chart, creates an unacceptable risk of crane overturning, and is a criminal liability. The lift must be stopped and re-planned: decrease the lift radius (boom in), improve ground bearing with outrigger pads, or bring in a larger crane.

Q50. Answer: A

A builder's level (optical or laser) establishes and verifies a horizontal datum plane across the site. It is used to set column base plate elevations, check beam levelness, and verify floor deck elevations.

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